

Lawrence Y Long

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

Primary Major: S.B. in Physics

May 2026

Secondary Major: S.B. in Computer Science, Economics, and Data Science

- Minor: Mathematics; Cumulative GPA: 5.0/5.0

Relevant Coursework: Biological Circuit Engineering Lab; Biochemistry; Biomaterials (G); Organic Chemistry I and II; Deep Learning (G); Reinforcement Learning (G); Algorithms; Statistical Physics; Quantum Physics I and II; Quantum Information Science (G); Linear Algebra; Differential Equations; Relativistic Quantum Field Theory (G); String Theory; (G: graduate-level)

EXPERIENCE

Bin Zhang Group, MIT Department of Chemistry

Cambridge, MA

Undergraduate Researcher, Supervised by Dr. Joseph Paggi

Jun 2025 – Present

- Optimized novel balancing method in computational processing pipeline for high-resolution Micro-C data and applied generalizability to multiple cell types (see *Publications*)
- Applied optimized pipeline and maximum entropy inversion simulations to study local chromatin folding patterns across the cell cycle; found novel result of preservation of locally extended regions associated with promoters and enhancers in mitosis

Feng Zhang Lab, Broad Institute of MIT and Harvard

Cambridge, MA

Undergraduate Researcher, Supervised by Dr. Guilhem Faure

Jun 2024 – Present

- Created TIGRFinder, a tool that detects and annotates the TIGR arrays of Tas proteins (see *Publications*); used TIGRFinder to characterize the diversity and evolution of TIGR arrays, hinting towards its native functions.
- Developed genomic mining pipeline for novel RNA-guided system discovery using ribosomal proteins as a seed; computationally discovered one novel system that is currently being investigated experimentally
- Implemented reinforcement fine-tuning of large language model for peptide bioactivity classification and peptide cleavage site as screening method for experimental testing.
- Engineering AI-agent system for inverse discovery, i.e. automating hypothesis generation, (multi)-omics datasets analyses, and interpretation of results.

Forest White Lab, Koch Institute for Integrative Cancer Research at MIT

Cambridge, MA

Undergraduate Researcher, Supervised by Dr. Ryuhyun Ahn

Jan 2023 – Feb 2024

- Developed R pipeline for cell type abundance deconvolution of bulk proteomics data
- Integrated multi-omics glioblastoma data from various brain regions over proteomics, phospho-proteomics, spatial metabolomics, single-cell transcriptomics, and H&E staining via R and Python, uncovering microenvironment-dependent neuron-tumor interaction signals
- Assisted with development of INSIGHT (cell-subpopulation phosphoproteomics protocol, see *Publications*), by optimizing the antigen retrieval protocol and performing experiments.

MIT Department of Economics

Cambridge, MA

Undergraduate Researcher supervised by Dr. Deivy Houeix

Sep 2022 – Jun 2023

- Contributed to project studying network effects of digital payment technology adoption among taxi drivers in Senegal, in partnership with Wave (a mobile money company); manipulated, cleaned, and analyzed massive databases of gross transaction data (2.5 million people) and merchant surveys to study technology adoption

PUBLICATIONS

- Xu, P.*; **Long, L.Y.***, Faure, G.; Zhu, A.; Kim, S.; Zilberzwige-Tal, S.; Quinones-Olvera, N.; Evgeniou, L.; Flam-Shepherd, D.; Macrae, R.; Zhang, F. (2026); "TIGR-Tas diversity reveals co-evolved architecture and RNA-guided transcriptional repression." *Submitted to Nature Microbiology*; (* equal contribution)

- Paggi, J.M.; **Long, L.Y.**, Zhang, B. (2026); "Euchromatin forms condensed domains with short active regions on the surface." *Accepted at Nature Genetics*.
- Ahn, R.; D'Souza, A.D.; **Long, L.Y.**; Cui, Y.; Burgenske, D.; Bakken, K.K.; Ott, L.L.; Carlson, B.L.; Zhou, G.; Yaron-Barir, T.M.; Kohale, I.N.; Whittaker, C.A.; Flower, C.T.; Wyckoff, J.; Tuma, A.; Johnson, J.L.; Sarkaria, J.; White, F.M. (2024). "Uncovering the signaling networks of disseminated glioblastoma cells *in vivo* with INSIGHT." *Accepted at Nature Communications*.
<https://www.biorxiv.org/content/10.1101/2024.12.04.626717v1.abstract>

POSTER PRESENTATIONS

- **Long, L.**; Faure, G.; Zhang, F. (April 2025). "Dissecting TIGR arrays: TIGRFinder uncovers RNA maturation signatures, array polarity, and structural organization." Presented at Bioengineering Undergraduate Research Symposium at Cambridge, MA.
- **Long, L.**; Ahn, R.; D'Angelo, F.; Stopka, S.; Baquer, G.; Oh, M.; Regan, M.; Vaubel, R.; Tuma, A.; Burgenske, D.; Sarkaria, J.; Parney, I.; Burns, T.; Iavarone, A.; Agar, N.; White, F. (April 2024). "Multi-omic analysis of primary and disseminated glioblastoma." Presented at Bioengineering Undergraduate Research Symposium at Cambridge, MA.

TECHNICAL SKILLS

- **Lab techniques:** flow cytometry; immunohistochemistry; mutagenesis; nucleic acid extraction, purification, and quantification; molecular cloning and primer design; protein purification; confocal microscopy
- **Programming languages:** Python, R, Bash
- **Computational skills:** deep learning, omics analysis – metagenomics, viromics, metatranscriptomics, RNAseq (sc, sn, bulk), (phospho)proteomics, genomics, Hi-C, (spatial) metabolomics, epigenomics, multiomics integration, Geneious, BLAST, HMMER, AlphaFold, PyMOL, High-performance computing (HPC) clusters, coarse-grained polymer simulation, statistical analysis

ADDITIONAL

Awards: *International Biology Olympiad* representing Team USA (2022; top 4 of ~10,000 students); *USA Biology Olympiad (USABO) National Finalist* (2021, 2022; top 20 of ~10,000 students); *National Merit Scholar* (top 0.5% of 1.6 million students, 2022); *American Invitational Math Examination Qualifier* (2017)

Volunteering: TA for USABO National Finals Training Camp 2023, where I set up experiments, gave lectures, and wrote the final theoretical exam for choosing Team USA for the International Biology Olympiad. Question writer for USABO Opens and Semifinal Exams from 2022-2024.

Teaching: tutoring high school students as part of Baology USABO Prep Course (2022-2026), focusing on developing problem-solving skills; mentored ~500 students across 4 years.